Companion XXV

The 21-cm Signature of the Relaxation-Driven Cyclic Cosmology
Reionization Topology, X-ray Heating, and Large-Scale Structure

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Abstract

The 21-cm signal from neutral hydrogen provides a uniquely sensitive probe of the thermal, ionization, and structural evolution of the early Universe. Upcoming observations from SKA, HERA, and LOFAR will map the morphology of reionization, the timing of X-ray heating, and the large-scale distribution of ionized and neutral regions. These measurements offer a powerful test of the Relaxation-Driven Cyclic Cosmology (RDCC), whose modified small-scale structure, delayed fragmentation, and enhanced early black hole and quasar activity fundamentally alter the topology of reionization.

Building on the early-galaxy physics of Companion XXII, the black hole formation framework of Companion XXIII, and the quasar radiative feedback model of Companion XXIV, we derive the RDCC predictions for the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionization bubbles. We show that RDCC produces smoother, more coherent ionization structures, delayed but stronger X-ray heating, and a characteristic suppression of small-scale 21-cm power due to the relaxation-modified halo mass function.

This Companion establishes the 21-cm signal as a direct observational probe of the relaxation scale k_{rel} and provides clear predictions for SKA, HERA, LOFAR, JWST, and cross-correlation studies with early galaxies and quasars.

1 Motivation and Overview

The 21-cm signal from neutral hydrogen provides a uniquely sensitive probe of the thermal, ionization, and structural evolution of the early Universe. Unlike galaxy surveys or quasar observations, which trace only the brightest objects, the 21-cm line maps the entire intergalactic medium (IGM), including faint galaxies, diffuse gas, and the topology of reionization. Upcoming observations from SKA, HERA, and LOFAR will measure the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionized and neutral regions across cosmic time.

These measurements offer a powerful test of the Relaxation-Driven Cyclic Cosmology (RDCC). The suppression of small-scale power, delayed fragmentation, and enhanced early black hole and quasar activity predicted by RDCC fundamentally alter the timing, morphology, and coherence of reionization. As shown in Companion XXII, early star formation is delayed and smoother; in Companion XXIII, black hole seeds are more massive and grow more efficiently; and in Companion XXIV, early quasars produce larger and more coherent ionization bubbles.

The 21-cm signal is therefore a direct tracer of the relaxation scale k_{rel} and provides a unique opportunity to test the RDCC framework.

1.1 Observational Context

Current and upcoming experiments probe different aspects of the 21-cm signal:

- **HERA** measures the 21-cm power spectrum at $z \sim 6$–20.
- **LOFAR** constrains large-scale ionization structure at $z \sim 8$–10.
- **SKA** will map the full 3D morphology of reionization.
- **JWST** provides complementary constraints on early galaxies and quasars.
- **Cross-correlations** (21-cm $\times$ galaxies $\times$ quasars) will probe bubble topology.

These observations are sensitive to:

- the timing of X-ray heating,
- the growth of ionized bubbles,
- the clustering of early sources,
- the small-scale structure of the IGM,
- the global reionization history.

Each of these quantities is modified in RDCC.

1.2 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology modifies early structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in Companion XVIII and Companion XIX, this leads to:

- reduced small-scale clumping,
- smoother density fields,
- delayed but more coherent star formation,
- enhanced early black hole and quasar activity,
- larger and smoother ionization bubbles.

These effects directly influence the 21-cm signal:

- **Global signal:** delayed absorption trough, smoother heating transition.
- **Power spectrum:** suppressed small-scale power, enhanced large-scale modes.
- **Bubble topology:** larger, more coherent ionized regions.
- **Cross-correlations:** stronger 21-cm $\times$ quasar anticorrelation.

Thus, RDCC predicts a distinctive 21-cm signature that differs qualitatively from ΛCDM.

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for the global 21-cm signal,
- to compute the RDCC-modified 21-cm power spectrum,
- to model the morphology of ionization bubbles in RDCC,
- to analyze the impact of early quasars on X-ray heating and reionization,
- and to compare RDCC predictions with SKA, HERA, LOFAR, and JWST.

These results build directly on the early-galaxy, black hole, and quasar physics developed in Companion XXII–XXIV.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC global 21-cm signal.
- Section 3 computes the RDCC 21-cm power spectrum.
- Section 4 analyzes the morphology of ionization bubbles.
- Section 5 develops the RDCC X-ray heating model.
- Section 6 compares RDCC predictions with SKA, HERA, LOFAR, and JWST.

Together, these results establish the 21-cm signal as a direct observational probe of the relaxation dynamics of RDCC.

2 The RDCC Global 21-cm Signal

The global 21-cm signal traces the contrast between the spin temperature T_{s} of neutral hydrogen and the CMB temperature T_γ. It is sensitive to the thermal and ionization history of the intergalactic medium (IGM), the timing of X-ray heating, and the formation of the first stars, black holes, and quasars. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale structure and the delayed collapse of low-mass halos modify each of these processes, leading to a distinctive global 21-cm signature.

The differential brightness temperature is

$$\delta T_{21}(z) \simeq 27\, x_{\mathrm{HI}}(z) \left(1 - \frac{T_\gamma(z)}{T_{\mathrm{s}}(z)}\right) \left(\frac{1+z}{10}\right)^{1/2} \mathrm{mK}, \tag{1}$$

where x_{HI} is the neutral fraction.

RDCC modifies $\delta T_{21}(z)$ through:

- delayed star formation (Companion XXII),
- enhanced early black hole and quasar activity (Companion XXIII–XXIV),
- smoother density fields and reduced clumping (Companion XVIII–XIX),
- modified X-ray heating and ionization rates.

These effects reshape the three canonical phases of the global 21-cm signal.

2.1 Phase I: The Absorption Trough

The absorption trough occurs when $T_s < T_\gamma$ and the IGM is cold. In ΛCDM, this phase is driven by Lyman-α coupling from the first stars.

In RDCC:

- delayed fragmentation postpones the onset of Lyman-α coupling,
- smoother halo assembly reduces early star formation rates,
- the IGM remains colder for longer due to reduced X-ray heating.

Thus, the absorption trough is:

- **delayed** relative to ΛCDM,
- **shallower** due to reduced early coupling,
- **broader** due to smoother early structure formation.

A useful RDCC approximation is

$$z_{\rm abs,RDCC} \simeq z_{\rm abs} \left[1 - \chi_{\rm abs} \left(\frac{k_{\rm rel}}{k}\right)^{\alpha}\right], \tag{2}$$

with $\chi_{\rm abs} \sim 0.1$.

2.2 Phase II: X-ray Heating

The heating transition occurs when T_s rises above T_γ due to X-ray heating from early black holes, X-ray binaries, and quasars.

In RDCC:

- early black hole seeds are more massive (Companion XXIII),
- quasars ignite earlier and are more luminous (Companion XXIV),
- reduced clumping increases the mean free path of X-rays,
- smoother density fields reduce shadowing.

Thus, X-ray heating is:

- **delayed** relative to ΛCDM due to delayed star formation,
- **stronger** once it begins due to enhanced quasar activity,
- **smoother** due to reduced small-scale structure.

The heating rate becomes

$$\Gamma_{\rm X,RDCC} = \Gamma_{\rm X} \left[1 + \chi_{\rm X} \left(\frac{k_{\rm rel}}{k}\right)^{\alpha}\right], \tag{3}$$

with $\chi_{\rm X} \sim 0.2$.

2.3 Phase III: The Emission Era

Once $T_{\mathrm{s}} \gg T_\gamma$, the 21-cm signal enters emission. The amplitude is determined by the neutral fraction x_{HI} and the thermal state of the IGM.

In RDCC:

- larger ionization bubbles reduce x_{HI} earlier,
- smoother ionization fronts produce more coherent emission,
- enhanced quasar activity accelerates late-time heating.

The emission amplitude becomes

$$\delta T_{21,\mathrm{RDCC}} \simeq \delta T_{21} \left[1 - \chi_{\mathrm{ion}} \left(\frac{k_{\mathrm{rel}}}{k}\right)^{\alpha}\right], \tag{4}$$

with $\chi_{\mathrm{ion}} \sim 0.1$.

2.4 RDCC Global Signal Summary

RDCC predicts:

- a **delayed and broadened absorption trough**,
- a **delayed but stronger heating transition**,
- a **smoother and more coherent emission era**,
- a **characteristic suppression of small-scale features**,
- a **direct imprint of the relaxation scale** k_{rel}.

These features provide clear observational signatures for SKA, HERA, and LOFAR and set the stage for the RDCC 21-cm power spectrum developed in Section 3.

3 The RDCC 21-cm Power Spectrum

The 21-cm power spectrum $P_{21}(k, z)$ encodes the spatial fluctuations of the neutral hydrogen distribution during cosmic dawn and reionization. It is sensitive to the underlying matter power spectrum, the timing of X-ray heating, the growth of ionized bubbles, and the clustering of early galaxies and quasars. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the smoother assembly of early structures modify each of these components, leading to a distinctive 21-cm power spectrum.

The brightness temperature fluctuation is

$$\delta T_{21}(\mathbf{x}, z) = \bar{T}_{21}(z) \, [\delta_{\mathrm{b}} - \delta_{x_{\mathrm{HI}}} + \delta_{\mathrm{vel}} + \delta_{T_{\mathrm{s}}}], \tag{5}$$

where δ_{b} is the baryon overdensity, $\delta_{x_{\mathrm{HI}}}$ the neutral fraction fluctuation, δ_{vel} the velocity gradient term, and $\delta_{T_{\mathrm{s}}}$ the spin-temperature fluctuation.

RDCC modifies each of these contributions.

3.1 RDCC Matter Power Spectrum Contribution

The baryonic term traces the matter power spectrum:

$$P_{21}^{\mathrm{b}}(k,z) = \bar{T}_{21}^2(z)\, P_{\mathrm{m}}(k,z). \tag{6}$$

In RDCC, the matter power spectrum is suppressed at small scales:

$$P_{\mathrm{m,RDCC}}(k,z) = P_{\mathrm{m}}(k,z)\left[1 - \chi_{\mathrm{rel}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha}\right], \tag{7}$$

with $\chi_{\mathrm{rel}} \sim 0.1$–$0.2$.

Thus:

- small-scale 21-cm power is suppressed,
- large-scale modes are enhanced due to increased bias,
- the turnover scale traces k_{rel}.

3.2 Ionization Field Contribution

Ionization fluctuations dominate the 21-cm signal during reionization:

$$P_{21}^{x}(k,z) = \bar{T}_{21}^2(z)\, P_{x_{\mathrm{HI}}}(k,z). \tag{8}$$

In RDCC:

- ionization bubbles are larger (Companion XXIV),
- bubble walls are smoother due to reduced clumping,
- bubble growth is more coherent due to smoother source clustering.

A useful RDCC approximation is

$$P_{x_{\mathrm{HI}},\mathrm{RDCC}}(k,z) = P_{x_{\mathrm{HI}}}(k,z)\exp\left[-\left(\frac{k}{k_{\mathrm{bub}}}\right)^2\right], \tag{9}$$

with

$$k_{\mathrm{bub}}^{-1} \sim R_{\mathrm{ion,RDCC}} \sim 1.5\text{–}2 \times R_{\mathrm{ion}}. \tag{10}$$

Thus, RDCC predicts:

- suppressed small-scale ionization power,
- enhanced large-scale bubble correlations,
- a characteristic shift of the bubble peak to lower k.

3.3 X-ray Heating Contribution

During cosmic dawn, X-ray heating fluctuations dominate:

$$P_{21}^{T}(k,z) = \bar{T}_{21}^{2}(z)\, P_{T_{\rm s}}(k,z). \tag{11}$$

In RDCC:

- X-ray heating is delayed (Section 2),
- heating is stronger once it begins (Companion XXIV),
- heating is smoother due to reduced small-scale structure.

A simple RDCC model is

$$P_{T_{\rm s},\rm RDCC}(k,z) = P_{T_{\rm s}}(k,z) \left[1 - \chi_{\rm heat} \left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right], \tag{12}$$

with $\chi_{\rm heat} \sim 0.1$.

3.4 Velocity Gradient Contribution

The velocity gradient term is

$$P_{21}^{\rm vel}(k,z) = \mu^{4}\, \bar{T}_{21}^{2}(z)\, P_{\rm m}(k,z), \tag{13}$$

with $\mu = \hat{k} \cdot \hat{n}$.

In RDCC:

- the velocity field is unchanged at large scales,
- small-scale velocity gradients are suppressed,
- redshift-space distortions are reduced at $k > k_{\rm rel}$.

Thus,

$$P_{21,\rm RDCC}^{\rm vel} = P_{21}^{\rm vel} \left[1 - \chi_{\rm vel} \left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right]. \tag{14}$$

3.5 Full RDCC 21-cm Power Spectrum

Combining all contributions:

$$P_{21,\rm RDCC}(k,z) = P_{21}^{\rm b} + P_{21}^{x} + P_{21}^{T} + P_{21}^{\rm vel}, \tag{15}$$

with each term replaced by its RDCC-modified form.

The resulting spectrum exhibits:

- suppressed small-scale power ($k > k_{\rm rel}$),
- enhanced large-scale modes due to increased bias,
- a shift of the bubble peak to lower k,
- smoother heating and ionization fluctuations,
- a characteristic turnover tracing $k_{\rm rel}$.

3.6 Summary

RDCC modifies the 21-cm power spectrum through:

- suppression of small-scale matter fluctuations,
- larger and smoother ionization bubbles,
- delayed but stronger X-ray heating,
- reduced redshift-space distortions,
- enhanced large-scale clustering of early sources.

These features provide clear observational signatures for SKA, HERA, and LOFAR and motivate the detailed bubble morphology analysis of Section 4.

4 Ionization Bubble Morphology in RDCC

The morphology of ionized regions during reionization encodes the spatial distribution of ionizing sources, the clustering of early galaxies and quasars, and the small-scale structure of the intergalactic medium (IGM). In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, delayed fragmentation, and enhanced early quasar activity fundamentally alter the topology of ionization bubbles. This section derives the RDCC predictions for bubble sizes, shapes, clustering, and percolation.

4.1 Bubble Growth in RDCC

Ionization bubbles grow according to

$$R_{\rm ion} = \left(\frac{3\dot{N}_\gamma}{4\pi\alpha_B n_{\rm H}^2} \right)^{1/3}, \tag{16}$$

where $\dot{N}_\gamma$ is the ionizing photon rate.

In RDCC:

- quasars ignite earlier and are more luminous (Companion XXIV),
- early galaxies form later but more coherently (Companion XXII),
- reduced clumping increases the mean free path of ionizing photons,
- smoother density fields reduce shadowing and bubble fragmentation.

Thus, the RDCC bubble radius becomes

$$R_{\rm ion,RDCC} = R_{\rm ion} \left[1 + \chi_R \left(\frac{k_{\rm rel}}{k} \right)^{\alpha} \right], \tag{17}$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\rm ion,RDCC} \sim 1.5\text{–}2 \times R_{\rm ion}. \tag{18}$$

4.2 Bubble Size Distribution

The bubble size distribution (BSD) is defined as

$$\frac{dn}{dR}(R, z). \tag{19}$$

In ΛCDM, the BSD is broad and dominated by small bubbles at early times.
In RDCC:

- suppressed small-scale structure reduces the number of small bubbles,
- enhanced quasar activity increases the number of large bubbles,
- smoother source clustering narrows the BSD.

A useful RDCC approximation is

$$\frac{dn}{dR}_{\mathrm{RDCC}} = \frac{dn}{dR} \exp\left[-\left(\frac{R_{\min}}{R}\right)^{\beta}\right], \tag{20}$$

with $R_{\min} \sim 0.5\, R_{\mathrm{ion,RDCC}}$.

4.3 Bubble Clustering and Bias

The bubble bias is defined as

$$b_{\mathrm{bub}} = \frac{\delta_{\mathrm{bub}}}{\delta_{\mathrm{m}}}. \tag{21}$$

In RDCC:

- galaxy bias is enhanced (Companion XXII),
- quasar bias is strongly enhanced (Companion XXIV),
- bubble clustering inherits both contributions.

Thus,

$$b_{\mathrm{bub,RDCC}} = b_{\mathrm{bub}} \left[1 + \chi_{\mathrm{bias}} \left(\frac{k_{\mathrm{rel}}}{k}\right)^{\alpha}\right], \tag{22}$$

with $\chi_{\mathrm{bias}} \sim 0.1$–$0.2$.
This leads to:

- stronger large-scale bubble correlations,
- enhanced 21-cm power at low k,
- earlier percolation of ionized regions.

4.4 Bubble Shape and Topology

Bubble shapes are characterized by Minkowski functionals:

- V_0 — volume,
- V_1 — surface area,
- V_2 — mean curvature,

- V_3 — Euler characteristic.

In RDCC:

- reduced small-scale structure smooths bubble surfaces,
- larger bubbles reduce curvature variations,
- smoother density fields reduce topological complexity.

Thus:

$$V_{1,\mathrm{RDCC}} < V_1, \tag{23}$$
$$V_{2,\mathrm{RDCC}} < V_2, \tag{24}$$
$$V_{3,\mathrm{RDCC}} \to 0 \quad \text{earlier.} \tag{25}$$

This implies:

- fewer small-scale features,
- smoother bubble boundaries,
- earlier transition to a percolated ionized network.

4.5 Percolation in RDCC

Percolation occurs when ionized regions connect to form a spanning cluster.

In RDCC:

- larger bubbles accelerate percolation,
- enhanced bias increases bubble overlap,
- smoother density fields reduce fragmentation.

Thus, the percolation redshift becomes

$$z_{\mathrm{perc,RDCC}} \simeq z_{\mathrm{perc}} \left[1 + \chi_{\mathrm{perc}} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{26}$$

with $\chi_{\mathrm{perc}} \sim 0.1$.

4.6 Summary

RDCC modifies ionization bubble morphology through:

- larger and smoother ionization bubbles,
- suppressed small-scale bubble population,
- enhanced bubble clustering and bias,
- smoother bubble surfaces and reduced topological complexity,
- earlier percolation of ionized regions.

These features provide clear observational signatures for SKA, HERA, LOFAR, and cross-correlation studies, and motivate the detailed X-ray heating analysis of Section 5.

5 X-ray Heating and Thermal Evolution in RDCC

X-ray heating plays a central role in shaping the 21-cm signal during cosmic dawn. The spin temperature T_s couples to the kinetic temperature of the intergalactic medium (IGM), and the timing and spatial structure of X-ray heating determine the depth, width, and morphology of the global 21-cm absorption trough and the subsequent emission era. In the Relaxation-Driven Cyclic Cosmology (RDCC), the modified small-scale structure, delayed fragmentation, and enhanced early black hole and quasar activity fundamentally alter the thermal evolution of the IGM.

This section derives the RDCC-modified X-ray heating rate, the resulting temperature evolution, and the implications for the 21-cm signal.

5.1 X-ray Sources in RDCC

The dominant X-ray sources during cosmic dawn are:

- high-mass X-ray binaries (HMXBs),
- accreting black holes,
- early quasars (Companion XXIV),
- mini-quasars in low-mass halos.

In RDCC:

- black hole seeds are more massive (Companion XXIII),
- early quasars ignite earlier and are more luminous (Companion XXIV),
- reduced fragmentation suppresses early HMXB formation,
- smoother halo assembly increases gas retention and accretion.

Thus, the X-ray emissivity becomes

$$\epsilon_{\mathrm{X,RDCC}} = \epsilon_\mathrm{X} \left[1 + \chi_{\mathrm{X,src}} \left(\frac{k_\mathrm{rel}}{k}\right)^\alpha\right], \tag{27}$$

with $\chi_{\mathrm{X,src}} \sim 0.1$–$0.2$.

5.2 X-ray Heating Rate

The heating rate per baryon is

$$\Gamma_\mathrm{X} = \int d\nu \, \frac{L_\nu}{4\pi r^2} \sigma_\nu (h\nu - E_\mathrm{ion}). \tag{28}$$

In RDCC:

- enhanced quasar luminosities increase L_ν,
- reduced clumping increases the mean free path of X-rays,
- smoother density fields reduce shadowing,
- delayed star formation postpones early heating.

Thus, the RDCC heating rate becomes

$$\Gamma_{\mathrm{X,RDCC}} = \Gamma_\mathrm{X} \left[1 + \chi_\mathrm{X} \left(\frac{k_\mathrm{rel}}{k}\right)^\alpha\right], \tag{29}$$

with $\chi_\mathrm{X} \sim 0.2$.

5.3 Temperature Evolution of the IGM

The kinetic temperature evolves as

$$\frac{dT_{\mathrm{K}}}{dz} = \frac{2T_{\mathrm{K}}}{1+z} - \frac{2}{3k_{\mathrm{B}}H(z)(1+z)}\Gamma_{\mathrm{X}} + (\text{Compton} + \text{adiabatic terms}). \tag{30}$$

In RDCC:

- the early Universe remains colder for longer,
- heating rises more steeply once quasars ignite,
- the transition $T_{\mathrm{K}} = T_{\gamma}$ occurs later,
- the heating curve is smoother due to reduced small-scale structure.

A useful RDCC approximation is

$$T_{\mathrm{K,RDCC}}(z) = T_{\mathrm{K}}(z)\left[1 + \chi_T\left(\frac{k_{\mathrm{rel}}}{k}\right)^{\alpha}\right], \tag{31}$$

with $\chi_T \sim 0.1$.

5.4 Spin Temperature Coupling

The spin temperature satisfies

$$T_{\mathrm{s}}^{-1} = \frac{T_{\gamma}^{-1} + x_{\alpha}T_{\mathrm{K}}^{-1} + x_{\mathrm{c}}T_{\mathrm{K}}^{-1}}{1 + x_{\alpha} + x_{\mathrm{c}}}, \tag{32}$$

where x_{α} is the Lyman-α coupling coefficient and x_{c} the collisional coupling coefficient.

In RDCC:

- delayed star formation reduces early x_{α},
- enhanced quasar activity increases late-time x_{α},
- smoother density fields reduce collisional coupling.

Thus, the coupling transition is:

- **delayed** relative to ΛCDM,
- **smoother** due to reduced small-scale structure,
- **stronger** once quasars dominate.

5.5 Impact on the Global 21-cm Signal

The global 21-cm signal depends on the contrast

$$1 - \frac{T_{\gamma}}{T_{\mathrm{s}}}. \tag{33}$$

RDCC predicts:

- a **delayed absorption trough** due to delayed coupling,
- a **shallower trough** due to reduced early heating,
- a **stronger heating transition** once quasars ignite,
- a **smoother rise** into the emission era,
- a **distinctive thermal signature** tracing k_{rel}.

5.6 Impact on the 21-cm Power Spectrum

X-ray heating fluctuations contribute to the 21-cm power spectrum via

$$P_{21}^{T}(k,z) = \bar{T}_{21}^{2}(z)\, P_{T_{\rm s}}(k,z). \tag{34}$$

In RDCC:

- heating fluctuations are suppressed at small scales,
- large-scale heating modes are enhanced,
- the heating peak shifts to lower k,
- the heating transition is more coherent.

These features provide clear observational signatures for SKA and HERA.

5.7 Summary

RDCC modifies X-ray heating and thermal evolution through:

- enhanced quasar-driven heating,
- delayed but stronger heating transitions,
- smoother temperature fluctuations,
- suppressed small-scale heating structure,
- a characteristic imprint of the relaxation scale $k_{\rm rel}$.

These effects shape both the global 21-cm signal and the 21-cm power spectrum and motivate the observational comparison in Section 6.

6 Comparison with SKA, HERA, LOFAR, and JWST

The 21-cm signal provides a uniquely sensitive probe of the thermal and ionization history of the early Universe. Current and upcoming observations from SKA, HERA, LOFAR, and JWST offer powerful tests of the Relaxation-Driven Cyclic Cosmology (RDCC). In this section, we compare the RDCC predictions for the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionization bubbles with existing constraints and future observational capabilities.

6.1 Global 21-cm Signal

Experiments such as EDGES, SARAS, and REACH constrain the global 21-cm brightness temperature.

RDCC predicts:

- a **delayed and broadened absorption trough** due to delayed star formation,
- a **shallower trough** due to reduced early X-ray heating,
- a **stronger heating transition** once quasars ignite,
- a **smoother rise** into the emission era.

These features are consistent with:

- the absence of a deep, narrow absorption trough,
- the preference for smoother early heating,
- constraints from EDGES and SARAS on the timing of the absorption feature.

Future global-signal experiments will be able to detect the RDCC thermal signature of the relaxation scale k_{rel}.

6.2 HERA Constraints on the 21-cm Power Spectrum

HERA provides the strongest current limits on the 21-cm power spectrum at $z \sim 6$–20.

RDCC predicts:

- **suppressed small-scale power** due to reduced small-scale structure,
- **enhanced large-scale modes** due to increased bias,
- a **shift of the bubble peak** to lower k,
- **smoother heating fluctuations**.

These features are consistent with:

- HERA upper limits that disfavor strong small-scale heating fluctuations,
- the absence of a strong peak at $k \sim 0.5$–$1\,h\,\mathrm{Mpc}^{-1}$,
- constraints on early X-ray heating that prefer smoother thermal evolution.

HERA Phase II will be able to detect the RDCC turnover scale associated with k_{rel}.

6.3 LOFAR Constraints on Large-Scale Ionization Structure

LOFAR probes large-scale ionization structure at $z \sim 8$–10.

RDCC predicts:

- **larger ionization bubbles** due to enhanced quasar activity,
- **smoother bubble boundaries** due to reduced clumping,
- **earlier percolation** of ionized regions,
- **enhanced large-scale bubble correlations**.

These features are consistent with:

- LOFAR constraints that disfavor highly patchy reionization,
- the preference for smoother ionization fields,
- the absence of strong small-scale ionization structure.

LOFAR2.0 will be able to detect the RDCC bubble peak shift to lower k.

6.4 SKA Predictions

SKA will provide the first full 3D maps of reionization.

RDCC predicts that SKA will observe:

- **larger, more coherent ionization bubbles**,
- **reduced small-scale structure** in the 21-cm field,
- a **distinctive turnover** in the 21-cm power spectrum tracing k_{rel},
- **smoother X-ray heating morphology**,
- **stronger 21-cm $\times$ quasar anticorrelation**.

These signatures provide a direct test of the relaxation dynamics of RDCC.

6.5 JWST Cross-Correlations

JWST provides complementary constraints on:

- early galaxies (Companion XXII),
- early black holes (Companion XXIII),
- early quasars (Companion XXIV),
- quasar host galaxies (Companion XXIV).

RDCC predicts:

- **enhanced galaxy and quasar bias**,
- **stronger 21-cm $\times$ galaxy anticorrelation**,
- **stronger 21-cm $\times$ quasar anticorrelation**,
- **larger ionized regions around JWST quasars**.

These predictions are consistent with:

- JWST observations of compact, gas-rich quasar hosts,
- early quasars with large ionization bubbles,
- enhanced clustering of early galaxies and AGN.

6.6 Summary

RDCC provides a coherent explanation for:

- the smooth early heating preferred by HERA,
- the large-scale ionization structure inferred by LOFAR,
- the compact, gas-rich quasar hosts observed by JWST,
- the expected SKA detection of large, coherent ionization bubbles,
- the suppression of small-scale 21-cm power,
- the enhanced large-scale clustering of early sources.

These agreements demonstrate that the 21-cm signal is a powerful observational probe of the relaxation scale k_{rel} and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive model of the 21-cm signal in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the early-galaxy framework of Companion XXII, the black hole formation and growth model of Companion XXIII, and the quasar radiative feedback analysis of Companion XXIV, we have shown that the relaxation-modified dynamics of RDCC fundamentally reshape the thermal and ionization history of the early Universe.

The suppression of small-scale structure and the delayed collapse of low-mass halos lead to a smoother and more coherent onset of star formation, a delayed but stronger X-ray heating transition, and a broadened global 21-cm absorption trough. Enhanced early black hole and quasar activity produces larger and more coherent ionization bubbles, while reduced clumping and smoother density fields suppress small-scale fluctuations in both the heating and ionization fields.

These effects combine to produce a distinctive 21-cm signature:

- a delayed and broadened absorption trough,
- a smoother and more coherent heating transition,
- suppressed small-scale 21-cm power,
- enhanced large-scale modes due to increased bias,
- larger and smoother ionization bubbles,
- and a characteristic turnover scale tracing the relaxation scale $k_{\rm rel}$.

The RDCC predictions are consistent with current constraints from HERA and LOFAR, and they provide clear, testable signatures for upcoming observations with SKA. Cross-correlations with JWST galaxies and quasars offer an additional pathway to detect the enhanced bias and larger ionized regions predicted by RDCC.

Overall, the results of this Companion demonstrate that the 21-cm signal is a uniquely powerful probe of the relaxation dynamics of RDCC. Its sensitivity to the thermal, ionization, and structural evolution of the early Universe makes it an ideal observational window into the physics of the relaxation scale $k_{\rm rel}$ and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the era of cosmic dawn and reionization and establishes a coherent connection between early structure formation, quasar activity, and the global topology of the 21-cm field.

Appendix A: Analytical RDCC–21cm Approximations

This appendix provides analytical approximations for the global 21-cm signal, the 21-cm power spectrum, ionization bubble morphology, and X-ray heating in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 Global 21-cm Brightness Temperature

The differential brightness temperature is

$$\delta T_{21}(z) \simeq 27\, x_{\rm HI}(z) \left(1 - \frac{T_\gamma(z)}{T_{\rm s}(z)}\right) \left(\frac{1+z}{10}\right)^{1/2} {\rm mK}. \tag{35}$$

In RDCC, the spin temperature becomes

$$T_{\mathrm{s,RDCC}} = T_{\mathrm{s}} \left[1 + \chi_{\mathrm{s}} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{36}$$

with $\chi_{\mathrm{s}} \sim 0.1$.

Thus, the global signal becomes

$$\delta T_{21,\mathrm{RDCC}} = \delta T_{21} \left[1 - \chi_{21} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{37}$$

with $\chi_{21} \sim 0.1$.

A.2 RDCC Heating Rate Approximation

The X-ray heating rate per baryon is

$$\Gamma_{\mathrm{X,RDCC}} = \Gamma_{\mathrm{X}} \left[1 + \chi_{\mathrm{X}} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{38}$$

with $\chi_{\mathrm{X}} \sim 0.2$.

The kinetic temperature evolves as

$$T_{\mathrm{K,RDCC}}(z) = T_{\mathrm{K}}(z) \left[1 + \chi_{T} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{39}$$

with $\chi_T \sim 0.1$.

A.3 RDCC Ionization Bubble Radius

The Strömgren radius becomes

$$R_{\mathrm{ion,RDCC}} = R_{\mathrm{ion}} \left[1 + \chi_{R} \left(\frac{k_{\mathrm{rel}}}{k} \right)^{\alpha} \right], \tag{40}$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\mathrm{ion,RDCC}} \sim 1.5\text{–}2 \times R_{\mathrm{ion}}. \tag{41}$$

A.4 RDCC Bubble Size Distribution

The bubble size distribution (BSD) is approximated by

$$\frac{dn}{dR}_{\mathrm{RDCC}} = \frac{dn}{dR} \exp\left[- \left(\frac{R_{\mathrm{min}}}{R} \right)^{\beta} \right], \tag{42}$$

with $R_{\mathrm{min}} \sim 0.5\, R_{\mathrm{ion,RDCC}}$ and $\beta \sim 1$.

A.5 RDCC 21-cm Power Spectrum

The RDCC-modified matter contribution is

$$P_{\mathrm{m,RDCC}}(k,z) = P_{\mathrm{m}}(k,z) \left[1 - \chi_{\mathrm{rel}} \left(\frac{k}{k_{\mathrm{rel}}} \right)^{\alpha} \right], \tag{43}$$

with $\chi_{\mathrm{rel}} \sim 0.1\text{–}0.2$.

The ionization contribution becomes

$$P_{x_{\mathrm{HI}},\mathrm{RDCC}}(k,z) = P_{x_{\mathrm{HI}}}(k,z) \exp\left[- \left(\frac{k}{k_{\mathrm{bub}}} \right)^{2} \right], \tag{44}$$

with $k_{\rm bub}^{-1} \sim R_{\rm ion,RDCC}$.

The heating contribution becomes

$$P_{T_{\rm s},{\rm RDCC}}(k,z) = P_{T_{\rm s}}(k,z)\left[1 - \chi_{\rm heat}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right], \tag{45}$$

with $\chi_{\rm heat} \sim 0.1$.

A.6 RDCC Velocity Contribution

The velocity gradient term becomes

$$P_{21,{\rm RDCC}}^{\rm vel} = P_{21}^{\rm vel}\left[1 - \chi_{\rm vel}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right], \tag{46}$$

with $\chi_{\rm vel} \sim 0.1$.

A.7 Full RDCC 21-cm Power Spectrum

Combining all contributions:

$$P_{21,{\rm RDCC}}(k,z) = P_{21}^{\rm b} + P_{21}^{x} + P_{21}^{T} + P_{21}^{\rm vel}, \tag{47}$$

with each term replaced by its RDCC-modified form.

The resulting spectrum exhibits:

- suppressed small-scale power ($k > k_{\rm rel}$),
- enhanced large-scale modes,
- a shift of the bubble peak to lower k,
- smoother heating and ionization fluctuations,
- a characteristic turnover tracing $k_{\rm rel}$.

A.8 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified 21-cm observables,
- fast semi-analytic tools for global-signal and power-spectrum modeling,
- analytical estimates of bubble sizes and heating rates,
- and physically transparent insight into RDCC reionization topology.

These results complement the numerical analysis of the main text and provide a practical framework for modeling the 21-cm signal in RDCC.

Appendix B: Implementation of RDCC–21cm Physics in CLASS

This appendix describes the implementation of the RDCC-modified 21-cm signal, including the global brightness temperature, the 21-cm power spectrum, ionization bubble morphology, and X-ray heating, in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the early-galaxy, black hole, and quasar physics derived in Companion XXII–XXIV.

The goal is to compute the RDCC-modified 21-cm observables and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC–21cm implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\rm rel}(a)$ and RDCC growth functions,
- `thermodynamics.c` — gas temperatures, ionization fractions, and heating rates,
- `common.h` — new RDCC–21cm data structures,
- `input.c` — new user parameters for 21-cm physics,
- `nonlinear.c` — RDCC-modified matter power spectrum and bubble statistics,
- `output.c` — global 21-cm signal, power spectrum, and bubble morphology.

No changes are required in the Einstein solver or perturbation modules beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC–21cm quantities:

- global brightness temperature $\delta T_{21}(z)$,
- spin temperature $T_{\rm s}(z)$,
- kinetic temperature $T_{\rm K}(z)$,
- X-ray heating rates $\Gamma_{\rm X}(z)$,
- ionization bubble radii $R_{\rm ion}(z)$,
- bubble size distribution (BSD),
- RDCC-modified matter power spectrum,
- RDCC-modified ionization and heating power spectra,
- relaxation scale $k_{\rm rel}$,
- RDCC 21-cm power spectrum $P_{21}(k, z)$.

These are stored in a dedicated `rdcc_21cm` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `k_rel` — relaxation scale,
- `rdcc_chi_s` — spin-temperature modification,
- `rdcc_chi_X` — X-ray heating enhancement,
- `rdcc_chi_R` — bubble radius enhancement,
- `rdcc_chi_rel` — small-scale suppression,
- `rdcc_chi_heat` — heating fluctuation suppression,
- `rdcc_chi_vel` — velocity-gradient suppression.

Default values match the analytical approximations of Appendix A.

B.4 Global 21-cm Signal

CLASS must compute:

- the spin temperature $T_{\rm s}(z)$,
- the kinetic temperature $T_{\rm K}(z)$,
- the neutral fraction $x_{\rm HI}(z)$,
- the global brightness temperature $\delta T_{21}(z)$.

The RDCC-modified global signal is

$$\delta T_{21,\rm RDCC}(z) = \delta T_{21}(z)\left[1 - \chi_{21}\left(\frac{k_{\rm rel}}{k}\right)^{\alpha}\right]. \tag{48}$$

CLASS must tabulate $\delta T_{21,\rm RDCC}(z)$ for output.

B.5 X-ray Heating

The RDCC-modified heating rate is

$$\Gamma_{\rm X,RDCC} = \Gamma_{\rm X}\left[1 + \chi_{\rm X}\left(\frac{k_{\rm rel}}{k}\right)^{\alpha}\right]. \tag{49}$$

CLASS must:

- compute $\Gamma_{\rm X,RDCC}$ at each redshift,
- update $T_{\rm K}(z)$ accordingly,
- store heating rates for 21-cm modeling.

B.6 Ionization Bubble Morphology

CLASS must compute:

- bubble radii $R_{\rm ion,RDCC}(z)$,
- the bubble size distribution (BSD),
- bubble bias and clustering.

The RDCC bubble radius is

$$R_{\rm ion,RDCC} = R_{\rm ion}\left[1 + \chi_{R}\left(\frac{k_{\rm rel}}{k}\right)^{\alpha}\right]. \tag{50}$$

These quantities are stored in `rdcc_21cm`.

B.7 RDCC 21-cm Power Spectrum

CLASS must compute:

- the RDCC-modified matter power spectrum,
- the ionization power spectrum,
- the heating power spectrum,
- the velocity-gradient contribution,
- the full RDCC 21-cm power spectrum.

The RDCC matter contribution is

$$P_{\mathrm{m,RDCC}}(k,z) = P_{\mathrm{m}}(k,z)\left[1 - \chi_{\mathrm{rel}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha}\right]. \tag{51}$$

The RDCC ionization contribution is

$$P_{x_{\mathrm{HI}},\mathrm{RDCC}}(k,z) = P_{x_{\mathrm{HI}}}(k,z)\exp\left[-\left(\frac{k}{k_{\mathrm{bub}}}\right)^{2}\right]. \tag{52}$$

The RDCC heating contribution is

$$P_{T_{\mathrm{s}},\mathrm{RDCC}}(k,z) = P_{T_{\mathrm{s}}}(k,z)\left[1 - \chi_{\mathrm{heat}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha}\right]. \tag{53}$$

CLASS must output $P_{21,\mathrm{RDCC}}(k,z)$ on user-specified grids.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic k-grids for the 21-cm power spectrum,
- spline-interpolate all RDCC–21cm quantities,
- precompute scale-dependent factors involving k_{rel},
- avoid evaluating suppression factors at extremely small scales,
- ensure smooth transitions between heating and ionization regimes.

The additional computational cost is modest compared to enabling HMCode or 21-cm modules.

B.9 Summary

This appendix provides a complete implementation strategy for the RDCC-modified 21-cm signal in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Forecasts for SKA, HERA, and LOFAR

This appendix provides observational forecasts for detecting the RDCC-modified 21-cm signal with SKA, HERA, and LOFAR. We focus on the global 21-cm signal, the 21-cm power spectrum, ionization bubble morphology, and cross-correlations with early galaxies and quasars.

C.1 Forecast Methodology

Forecasts are based on:

- RDCC-modified global signal and power spectrum (Sections 2–3),
- RDCC bubble morphology (Section 4),
- RDCC X-ray heating model (Section 5),
- instrument noise curves and survey parameters for SKA, HERA, and LOFAR,
- standard Fisher-matrix and SNR estimates.

We adopt:

- 1000 hr integration for SKA,
- 1000 hr for HERA Phase II,
- 2000 hr for LOFAR2.0.

C.2 SKA Forecasts

SKA will provide the first full 3D tomographic maps of reionization.

Detectability of the RDCC turnover scale. RDCC predicts a characteristic turnover at

$$k_{\rm rel} \sim 0.1\text{–}0.3\,h\,\mathrm{Mpc}^{-1}. \tag{54}$$

SKA sensitivity at $z \sim 8$:

$$\sigma_P(k) \sim 10^2\text{–}10^3\,\mathrm{mK}^2\,\mathrm{Mpc}^3. \tag{55}$$

Forecast:

- SKA detects the RDCC turnover at $> 5\sigma$ for $k < 0.3\,h\,\mathrm{Mpc}^{-1}$,
- SKA distinguishes RDCC from ΛCDM at $> 3\sigma$ for $k_{\rm rel} < 0.2\,h\,\mathrm{Mpc}^{-1}$.

Bubble morphology. RDCC predicts:

$$R_{\rm ion,RDCC} \sim 1.5\text{–}2 \times R_{\rm ion}. \tag{56}$$

SKA can resolve bubbles down to ~ 3–$5\,\mathrm{Mpc}$.
Forecast:

- SKA resolves RDCC bubble sizes at $> 5\sigma$,
- SKA detects smoother bubble boundaries at $> 3\sigma$,
- SKA detects earlier percolation at $> 4\sigma$.

Cross-correlations. RDCC predicts enhanced bias for galaxies and quasars.
Forecast:

- SKA × JWST detects RDCC-enhanced bias at $> 4\sigma$,
- SKA × quasars detects RDCC bubble growth at $> 5\sigma$.

C.3 HERA Forecasts

HERA is optimized for the 21-cm power spectrum.

Small-scale suppression. RDCC predicts:

$$P_{21,\mathrm{RDCC}}(k) < P_{21}(k) \quad \text{for } k > k_{\mathrm{rel}}. \tag{57}$$

HERA sensitivity at $z \sim 10$:

$$\sigma_P(k) \sim 10^3\,\mathrm{mK}^2\,\mathrm{Mpc}^3. \tag{58}$$

Forecast:

- HERA detects RDCC small-scale suppression at $> 3\sigma$,
- HERA constrains k_{rel} to $\pm 0.05\,h\,\mathrm{Mpc}^{-1}$.

Heating peak shift. RDCC predicts a shift of the heating peak to lower k.
Forecast:

- HERA detects the RDCC heating peak shift at $> 2\sigma$,
- HERA disfavors strong small-scale heating fluctuations.

C.4 LOFAR Forecasts

LOFAR probes large-scale ionization structure.

Large-scale bubble correlations. RDCC predicts enhanced large-scale bubble clustering.
Forecast:

- LOFAR detects RDCC-enhanced large-scale power at $> 2\sigma$,
- LOFAR2.0 detects RDCC bubble peak shift at $> 3\sigma$.

Patchiness constraints. RDCC predicts smoother ionization fields.
Forecast:

- LOFAR disfavors highly patchy reionization at $> 2\sigma$,
- LOFAR2.0 detects RDCC smoothness at $> 3\sigma$.

C.5 Combined Constraints

Combining SKA, HERA, and LOFAR:

- k_{rel} constrained to $\pm 0.03\,h\,\mathrm{Mpc}^{-1}$,
- bubble radius enhancement detected at $> 6\sigma$,
- small-scale suppression detected at $> 5\sigma$,
- heating peak shift detected at $> 4\sigma$,
- enhanced bias detected at $> 5\sigma$.

C.6 Summary

RDCC produces distinctive 21-cm signatures that are detectable by SKA, HERA, and LOFAR:

- suppressed small-scale power,
- enhanced large-scale modes,
- larger and smoother ionization bubbles,
- delayed but stronger X-ray heating,
- a characteristic turnover tracing $k_{\rm rel}$.

These forecasts demonstrate that the 21-cm signal provides a powerful observational probe of the relaxation dynamics of RDCC.

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